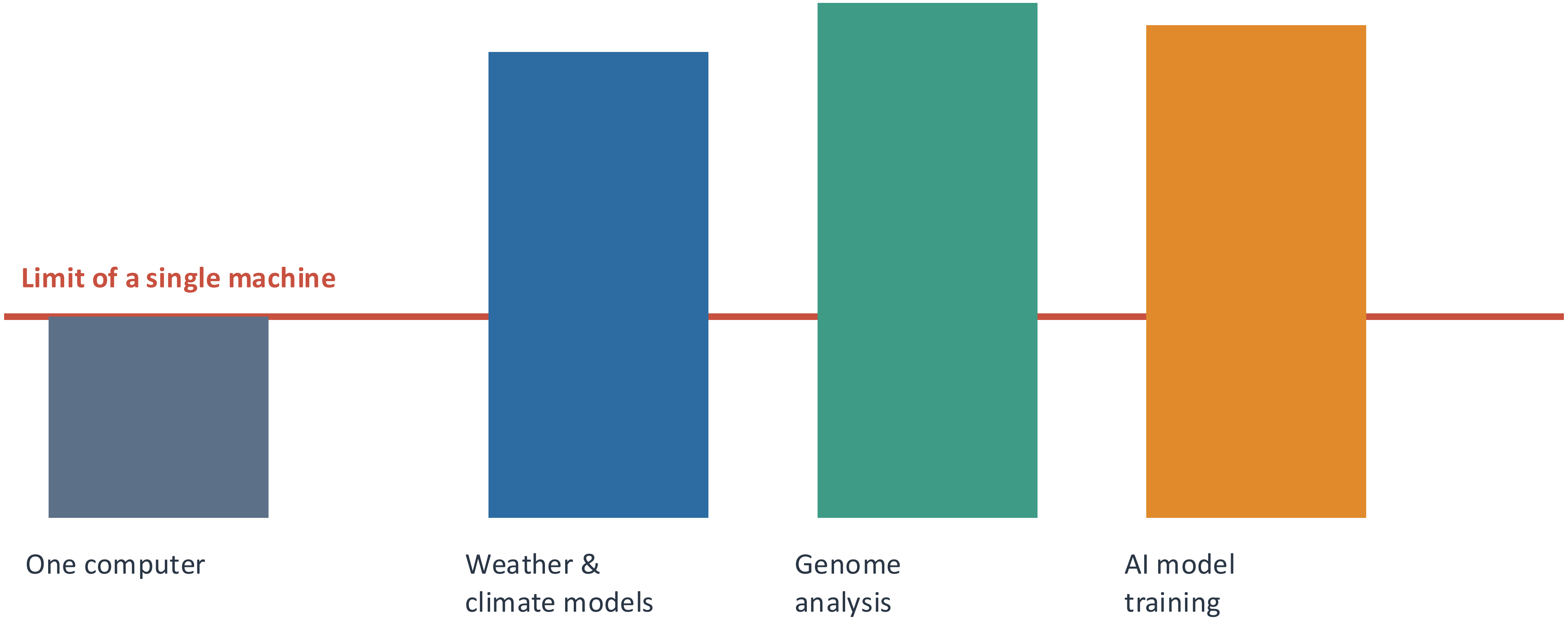
Understanding HPC Infrastructure

How compute, networking, storage and scheduling work together at scale

Some Problems Are Too Big for One Computer



Real workloads demand far more memory, compute, and time than any single machine can offer.

When One Server Is Not Enough

Some workloads exceed the compute power, memory, or practical runtime of a single server — no matter how large that server is.

Three limits that force scale-out: Compute capacity · Memory size · Time to result

Engineering & Simulation

Crash, airflow & stress models

Weather & Climate

Forecasts run against the clock

Healthcare & Genomics

Sequencing & analysis at scale

Large-Scale Analytics

Datasets too big for one node

AI Model Training

Billions of parameters

Decades before the AI boom, HPC was — and still is — the engine behind modern science and industry.

Parallelism in One Minute

1 WORKER — sequential

100 tasks, done one after another

each task starts only when the previous one finishes

≈ 100 units of time

10 WORKERS — parallel

the 100 tasks are divided among the workers

≈ 10 units of time *

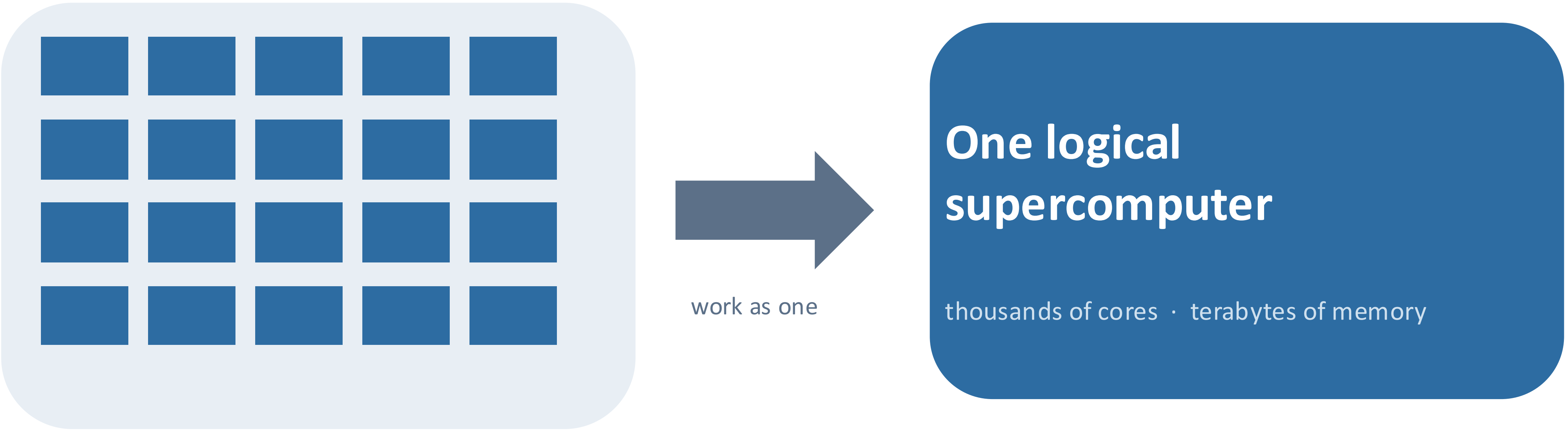
* only when the work can actually be split

Independent work
"Embarrassingly parallel." Pieces don't depend on each other, so adding workers gives near-linear speed-up.

Tightly coupled work
Workers must communicate and synchronize as they go — speed-up is capped by that coordination.

Coordination is overhead
Some workloads scale to thousands of workers; others barely speed up at all.

The Idea: Many Computers Acting as One



Many independent computers (nodes/servers)

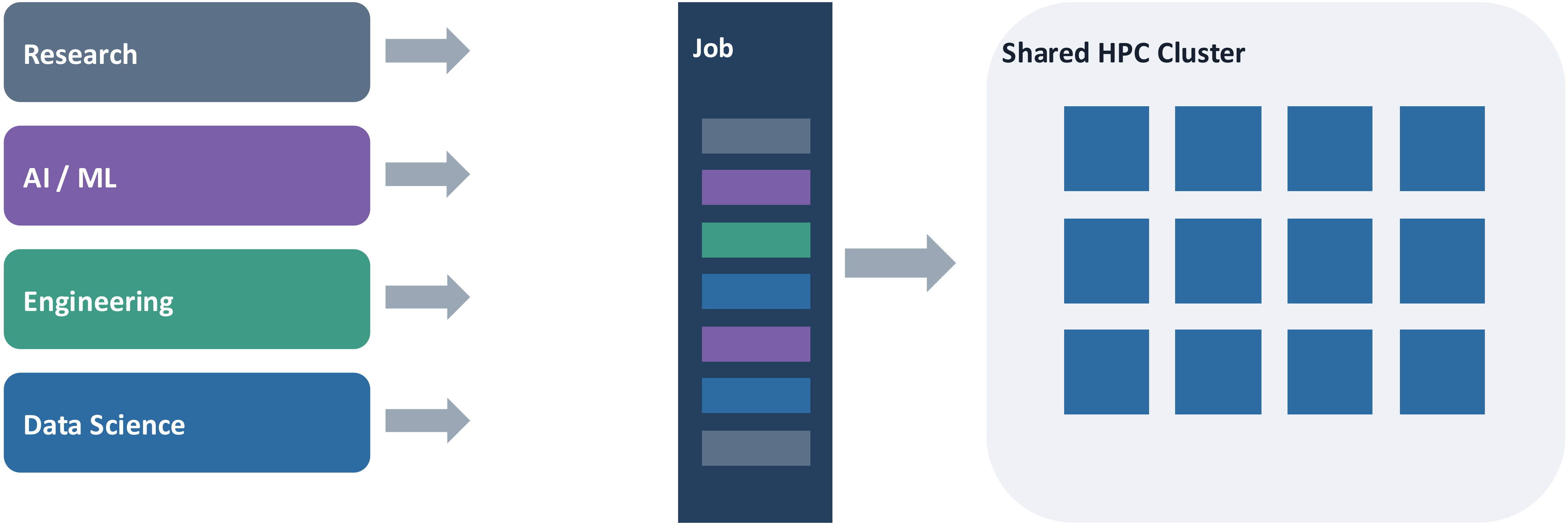
Tie enough machines together with the right network and software, and they behave as a single, far larger computer.

Main HPC Building Blocks

Each part has one clear role.

Login Node	**The front door** Where users connect, prepare their work, and submit jobs.
Head / Management Node	**The control room** Keeps the cluster organized and runs important management services.
Scheduler	**Air-traffic control** Decides which job runs, where it runs, and when it starts.
Compute Nodes	**The workforce** The servers that perform the actual calculations using CPUs, GPUs, or both.
Network	**The road system** Ethernet is the service road for everyday cluster traffic, while the high-speed network is the fast highway for performance-critical compute and storage traffic.
Shared Storage	**The shared workspace** Keeps files, datasets, and results available to the cluster. It may be NAS/NFS for everyday files or a parallel filesystem for large, high-speed workloads.

A Shared Supercomputer Needs Rules



Far more work is requested than the cluster can run at once — something must decide the order.

Scheduling: Fairness, Priority & Utilization

One cluster, shared capacity

Fair share

Over time, each team gets its agreed slice of the machine.

Priority

Urgent or high-value jobs can move up the queue.

Limits & quotas

No single user can take over the whole cluster.

Beyond fair-share, priority and quotas, the scheduler also handles queuing, placement, reservations and backfilling — balancing utilization, fairness and policy, not maximizing any single one.

Following One Job

One job, end to end - and which part of the cluster is responsible at each step.

EXAMPLE JOB REQUEST

4 nodes
32 CPU cores per node
or 4 GPUs per node
256 GB memory per node
2-hour maximum runtime

1. **Login** — Connect to a login node
2. **Prepare** — Stage data, write job script
3. **Submit** — Send job to the scheduler
4. **Queue** — Wait for resources to free
5. **Allocate** — Scheduler assigns nodes
6. **Execute** — Compute nodes run the job
7. **Communicate / R-W** — Nodes exchange data; read & write storage
8. **Finish** — Results saved, nodes released

Login node | Scheduler | Compute nodes | Network & storage

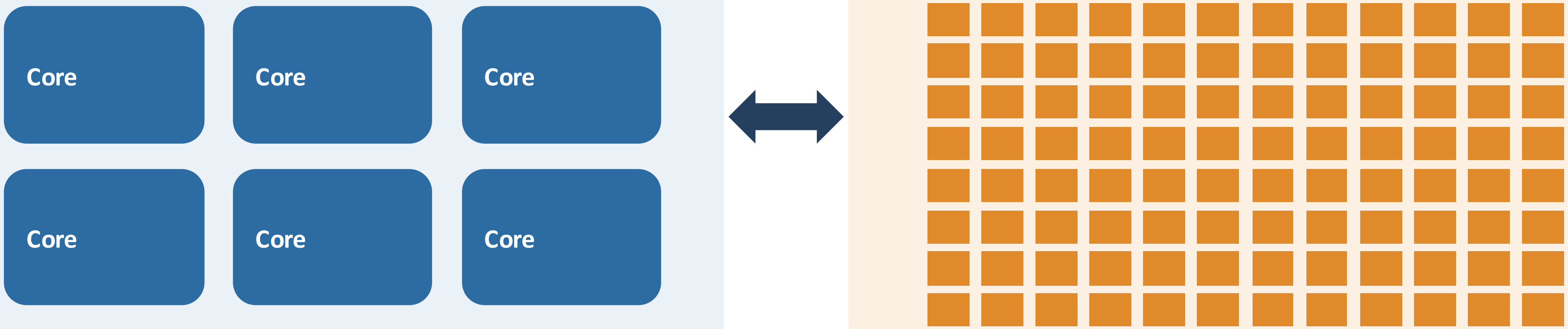
How CPUs and GPUs Work Together

CPU

Runs the application and handles general-purpose work

GPU

Performs many calculations in parallel



Prepares tasks, manages data movement, and sends suitable calculations to the GPU.

Executes heavy numerical work using many processing units at the same time.

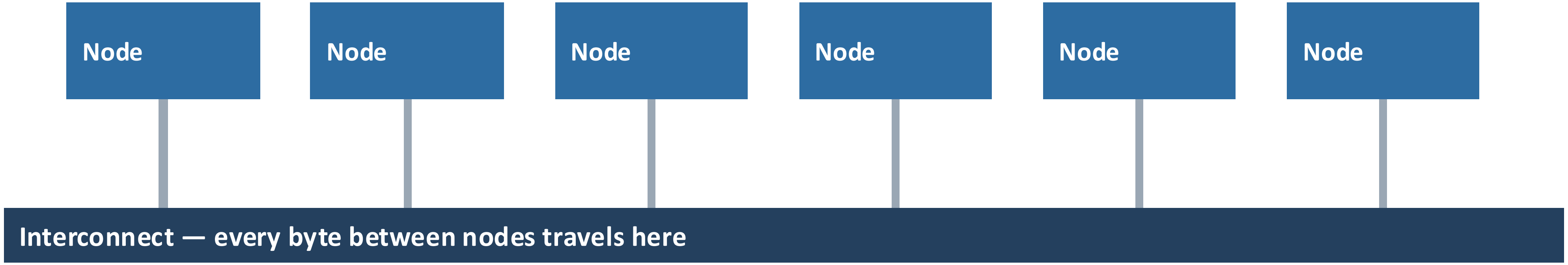
The CPU runs the overall application, while the GPU accelerates the highly parallel parts of the workload.

Faster Compute Exposed Every Other Bottleneck

Before GPUs	**Compute** the limit	**Network** keeps up	**Storage** keeps up
make compute faster ↓			
With GPUs	**Compute** now fast	**Network** new limit	**Storage** new limit

The bottleneck never disappears — it moves. Faster compute made the network and storage the new limits.

The Network Is the Real Backbone



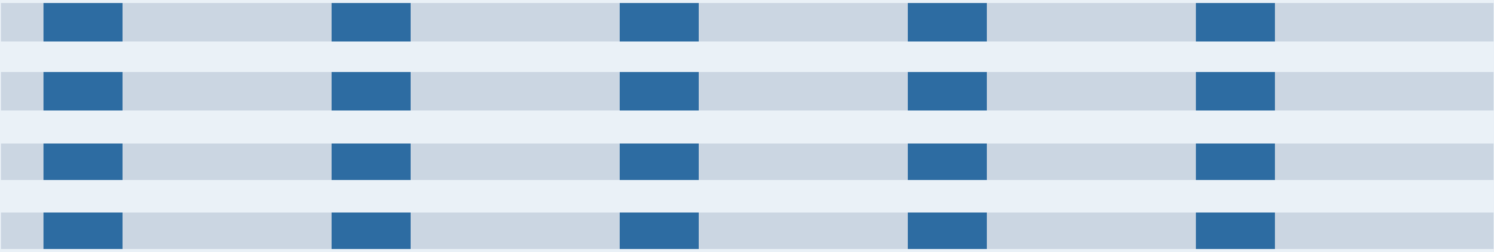
A fast processor is useless if it is always waiting for data.

At scale, performance may be limited by compute, memory, network, storage I/O, or software efficiency. Tightly coupled jobs are especially sensitive to network latency and congestion.

Latency vs. Bandwidth

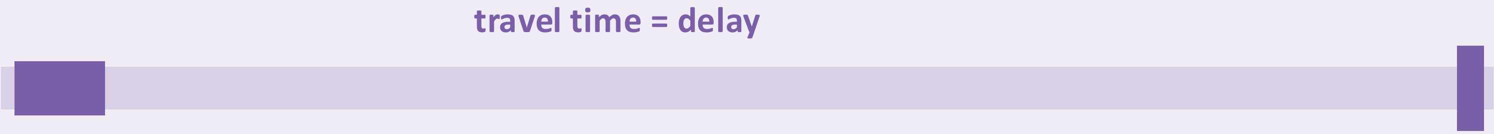
BANDWIDTH

how MUCH data moves at once



LATENCY

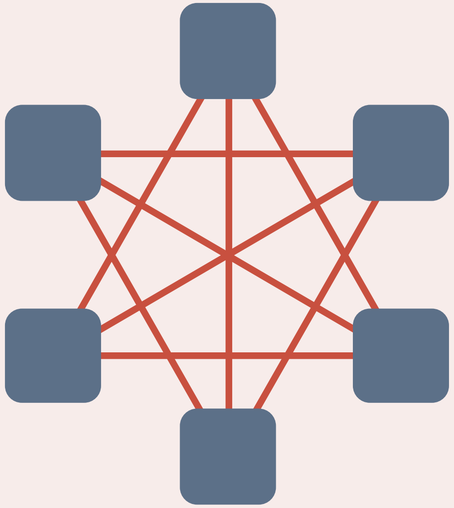
how SOON the first byte arrives



You need both — wide lanes (bandwidth) and a short trip (latency). HPC networks are engineered for both.

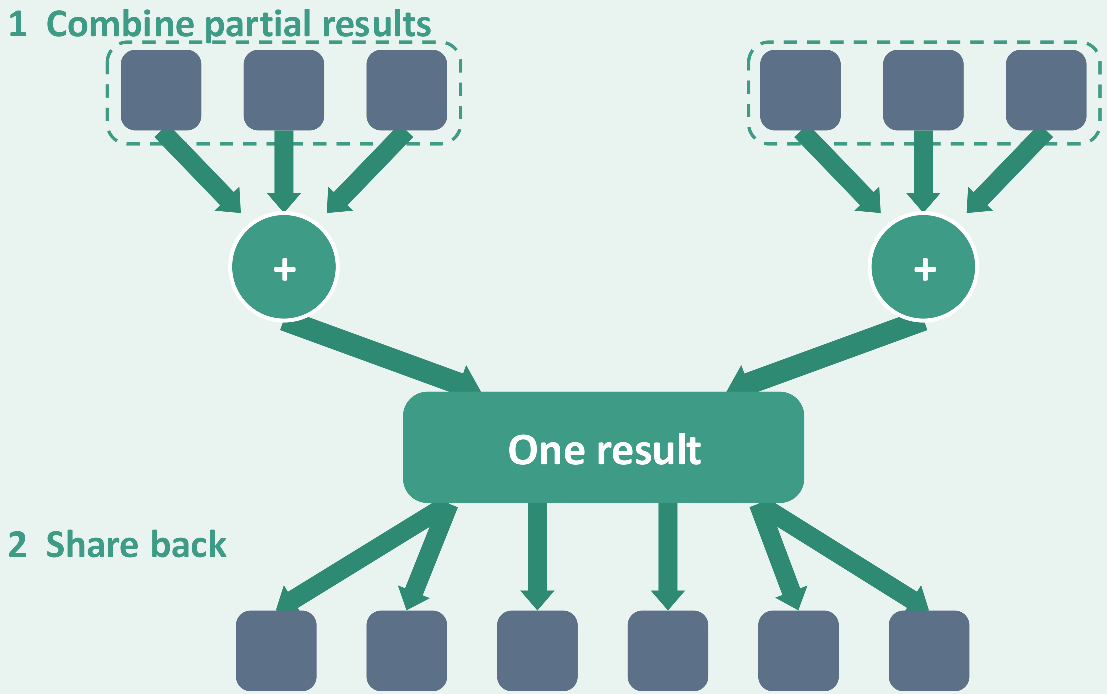
How Compute Nodes Communicate

Uncoordinated communication



Every node sends data directly to many others — traffic grows quickly.

Coordinated collective communication

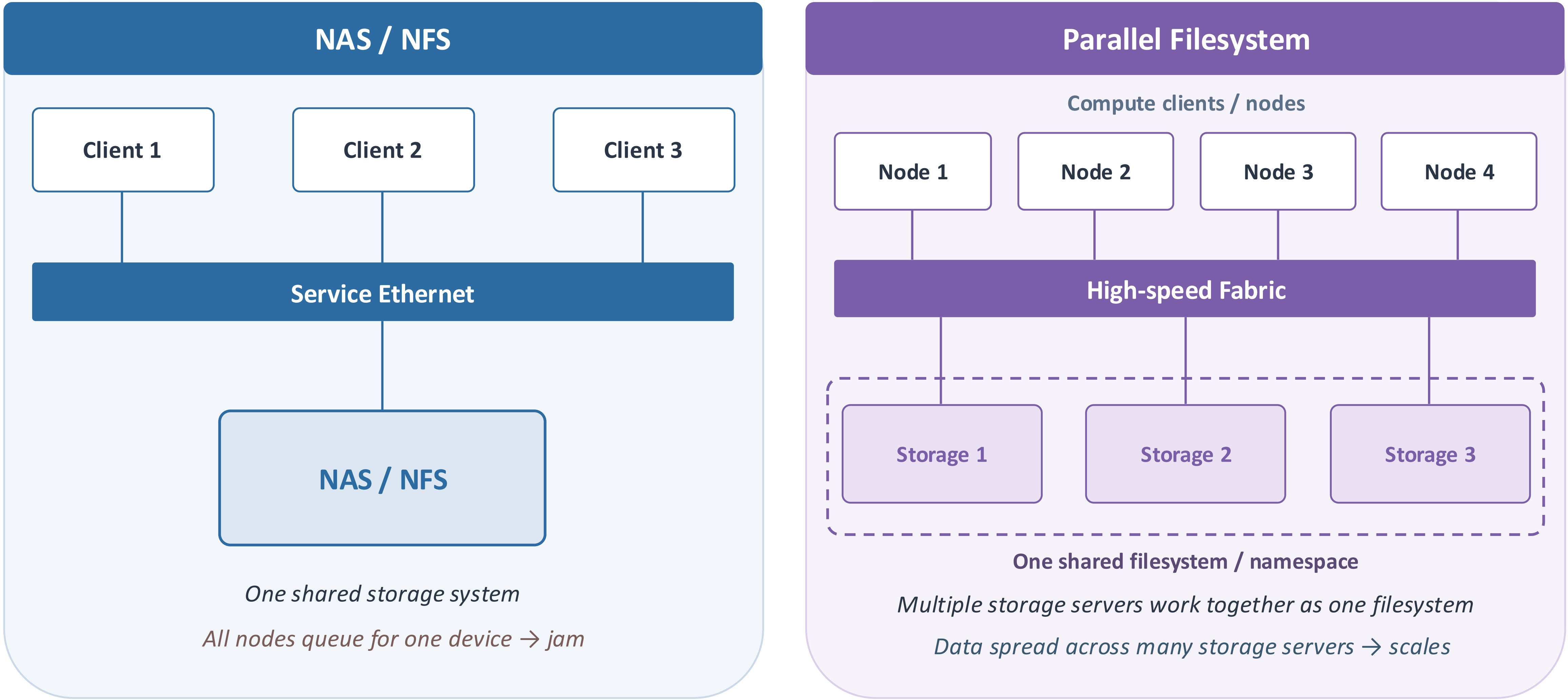


Partial results are combined into one result and then shared back to all nodes.

Structured communication reduces unnecessary traffic and helps the cluster scale better.

Shared Storage in HPC — NAS / NFS vs Parallel Filesystem

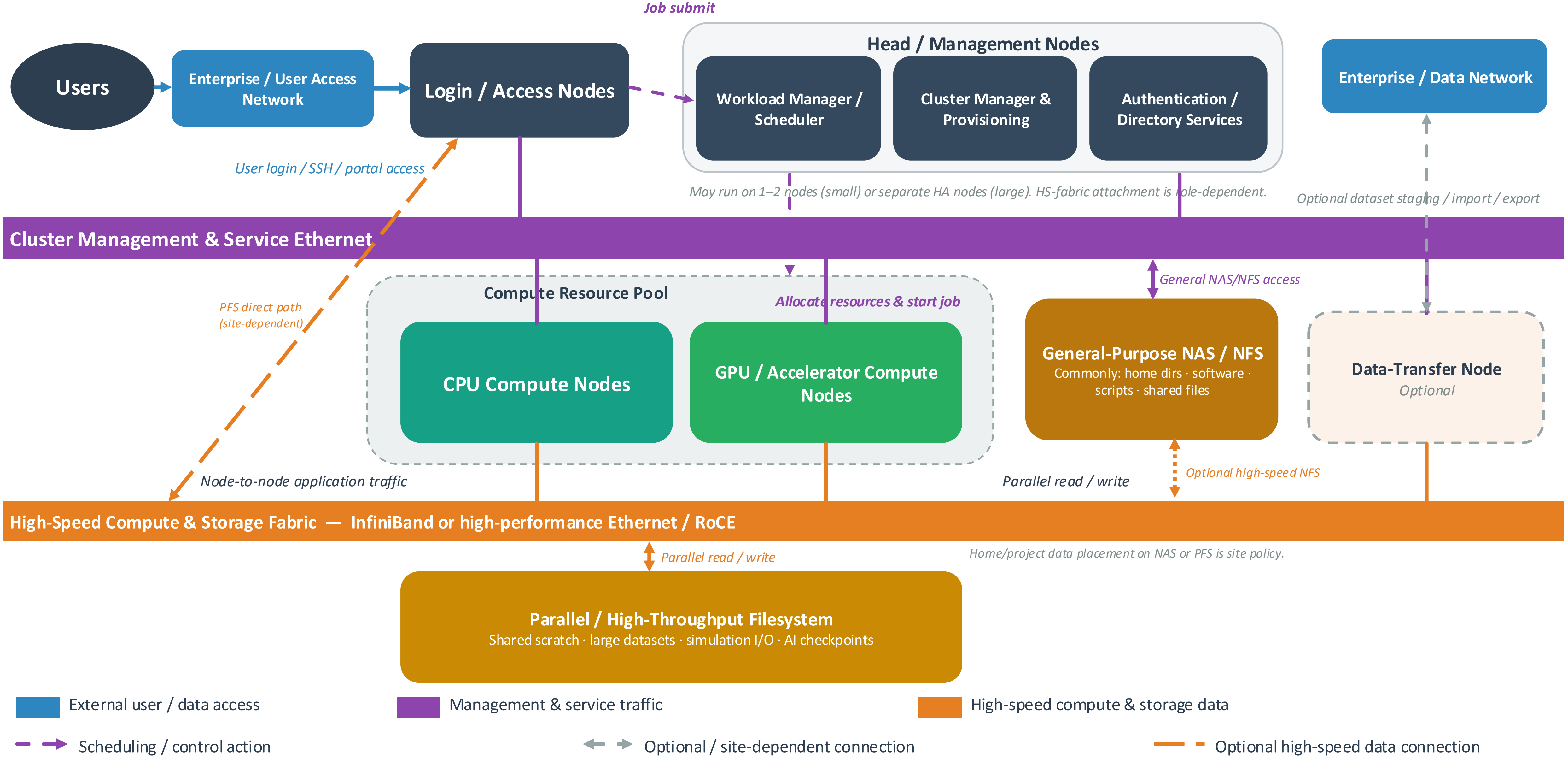
Both are shared storage, but they are designed for different kinds of access.



NAS / NFS: one general shared access path

Parallel filesystem: many storage servers presenting one shared filesystem

Modern HPC Architecture Overview



Logical reference architecture — services may be consolidated or separated; topology varies by cluster size, performance and site design. Out-of-band BMC / hardware-management networks omitted for clarity.

Same Building Blocks: Scientific HPC and AI Training Cluster

Scientific HPC Cluster	Infrastructure role	AI Training Cluster
Login node	Access	Login + control plane
Slurm scheduler	Scheduler	Slurm / Kubernetes
CPU + GPU nodes	Compute	GPU servers
GPU interconnect + scale-out fabric	Network	GPU interconnect + scale-out fabric
Parallel filesystem / shared storage	Storage	Parallel / object storage for datasets & checkpoints

Scientific HPC and large-scale AI share the same architectural principles — but their applications, software frameworks, data patterns and operational needs can differ.

Where AI Pushed HPC Further

Massive scale

From hundreds of nodes to tens of thousands of GPUs working on a single job.

Failure is routine

At that scale, parts fail mid-run — frequent checkpoints let jobs resume, not restart.

Compute-network co-design

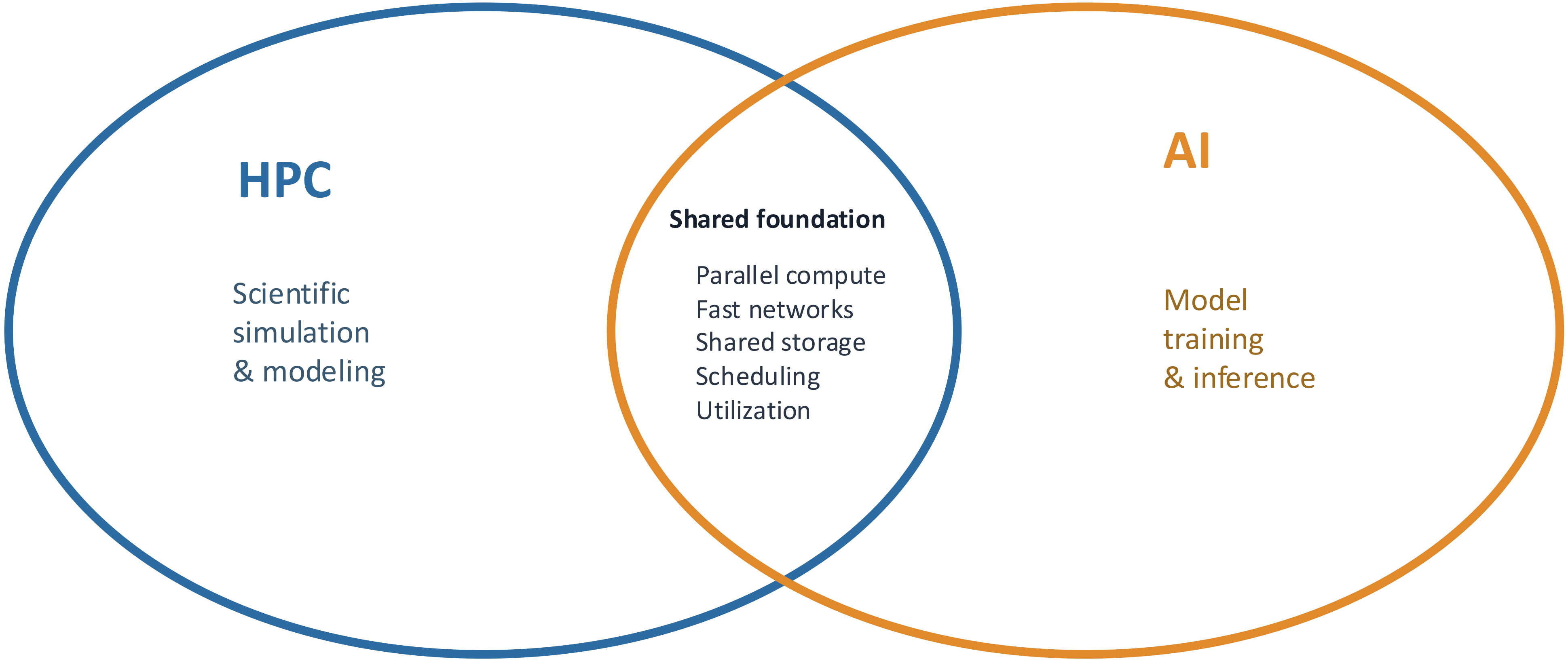
Interconnects like NVLink are now designed together with the chips themselves.

Always-on economics

Clusters run flat-out for months — idle time is simply unaffordable.

AI took HPC's principles to a scale HPC itself had rarely reached — and made them mainstream.

HPC and AI Are Converging



Two fields, once separate, are merging into a single large-scale computing practice.

What to Remember

1. Many machines, made to act as one — that is HPC.
2. Five principles power them all: parallel compute, fast networks, shared storage, scheduling, and utilization.
3. Modern AI is built on these very principles — not in spite of HPC, but because of it.

HPC isn't disappearing — it's the foundation modern AI is built on.

Thank You

HPC — the foundation behind modern large-scale computing and AI.